

/Users/halpern/teaching/352/labs/manual/master/rev_2022/skin.tex

Electromagnetic Skin Depth of Metals

1 Purpose

1. To understand and familiarize yourself with the occurrence and principles of electromagnetic shielding.
2. To quantitatively examine the skin depth problem by determining the skin depths of some metallic pipes.

2 Theory

It is common knowledge that electromagnetic radiation does not pass easily through a metal. The time-varying electric and magnetic fields generate relatively large screening currents in the metal; the characteristic length scale of the fields' resulting attenuation is the skin depth. This shielding “efficiency” is strongly dependent on the frequency of the radiation, as determined by the Maxwell equations. <https://www.overleaf.com/project/5c33d2446b7c1b7271687039> Let a long metal pipe be immersed in a uniform magnetic field which is varying with time as $e^{-i\omega t}$. The magnetic field is parallel to the cylinder axis, and this direction is taken to be the $\hat{z}$ direction of a cylindrical co-ordinate system. The pipe has an outer diameter $2R_2$ and inner diameter $2R_1$.

The amplitude and phase of the magnetic field inside the pipe can be calculated using Maxwell's equations. Our situation has cylindrical symmetry, so $\vec{E}$ and $\vec{H}$ depend only upon the radius. Moreover, $\vec{H}$ has only one component, H_z , and this component does not depend on z – how good is this approximation? Applying boundary conditions at the inner and outer surfaces of the pipe and taking the conductivity σ to be non-zero, an exact solution may be found to compute the attenuation and phase difference for the fields inside and outside the pipe. Due to the cylindrical nature of the problem, the solution involves Bessel's functions.

The solution can be simplified at high frequencies because Bessel's functions have rather simple limiting forms for large arguments. This solution is:

$$\frac{H_i}{H_o} = \rho e^{i\phi} \quad (1)$$

$$\rho = 2\sqrt{\frac{R_2}{R_1}} \left\{ \frac{e^{-k_o(R_2-R_1)}}{\sqrt{1 + R_1 k_o + \frac{R_1^2 k_o^2}{2}}} \right\} \quad (2)$$

$$\phi = k_o(R_2 - R_1) + \arctan\left(\frac{R_1 k_o}{2 + R_1 k_o}\right) \quad (3)$$

$$k_o = \sqrt{\frac{\omega \sigma \mu}{2}} \quad (4)$$

where

H_o = magnetic field outside the pipe

H_i = magnetic field inside the pipe

R_1 = inner radius of the pipe

R_2 = outer radius of the pipe

σ = conductivity of the metal ($\approx 3 \times 10^5 \frac{1}{\Omega \cdot \text{cm}}$ for Al)

μ = permeability of the metal

ω = angular frequency

★NOTE: This high frequency approximation breaks down for $k_o(R_2 - R_1) \lesssim 1$. You should check what frequency this is. It is recommended that you still take data below this point, to see how seriously the equations' breakdown is, and to check where it occurs.

3 Procedure

In this experiment you will measure the phase and amplitude of an alternating magnetic field inside a metal pipe relative to the field without the pipe. The schematic diagram of the apparatus is shown in figure 1. A Physics 409 lab at the same station uses steel pipes as well as aluminum and copper, so make sure that your aluminum pipe is really aluminum – there are some very significant differences between aluminum and steel!

A coil approximately 6" long, 2" in diameter, and wound with approximately 10 turns per cm, is connected to an HP3324A synthesized function generator through a 50Ω resistor to avoid short circuiting the output from the oscillator. This system is used to generate an AC magnetic field over the frequency range 100 Hz to 80 kHz. A pick-up coil consisting of 200 turns of #38 Formex insulated copper wire wound on a $\frac{3}{8}$ " diameter form is used to probe the magnetic field generated in the primary coil. The small e.m.f. generated in the pick-up

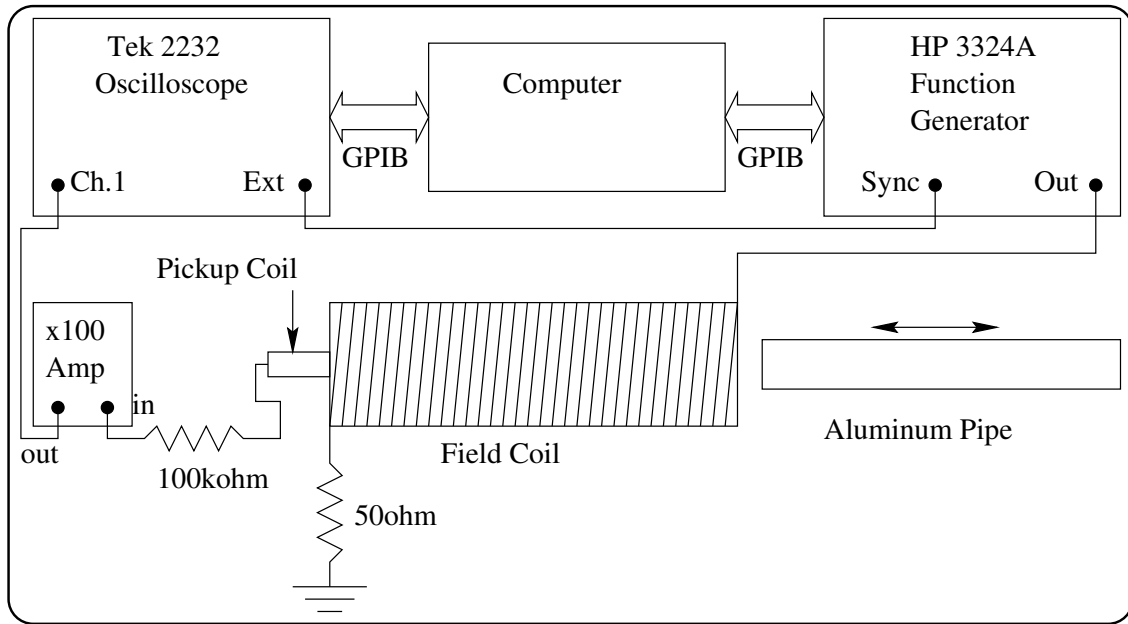


Figure 1: A schematic diagram of the skin depth apparatus.

coil is first amplified by a pre-amp with a $100\text{k}\Omega$ termination, then displayed on a Tek 2232 digital storage scope.

A LabView-based program on the computer's desktop is used to input the data from the scope to the PC. It will guide you through the data collection process and will control the function generator; you must control the oscilloscope. The program obtains a digitized version of the oscilloscope trace (1024 points), then fits it to a sinusoid of the form $A + B \cos(\omega t - \phi)$. For the most accurate fit, the voltage settings on the oscilloscope should keep the waveform as large as possible and there should be between 2 and 5 waveform periods visible. The program will prompt you to change the time scale on the oscilloscope between data points. The program outputs a tab-separated text file (*.SKD) with the frequency and fitting parameters (A , B and ϕ). It can also output waveform files (the digitized scope traces) if you want to check the fit.

The phase is measured from the nearest peak to the leftmost edge of the waveform image, so there is some ambiguity about what "phase" precisely means. While the data acquisition program is working, plot out the equation for ϕ as a function of ω with MATLAB and compare the result with what you see emerging from the experiment.

1. Open the Skin Depth program and follow the equipment setup instructions.
2. Using calipers, accurately measure the inner and outer radii of the two pipes. Using an accepted value for each conductivity, determine at what point the high-frequency approximation should break down.

3. Take data at frequencies between 100 Hz and 80 kHz, ensuring that the signal is not clipped or distorted by the pre-amp. Measure the amplitude and phase at each frequency for aluminum, copper, and air. You may need to re-take some of the high-frequency pipe data with a higher-amplitude signal from the function generator if the signal devolves into noise. This is easily corrected for later, especially if there's overlap in the data. Note that the program only attempts to save your data once, so save it to the hard disk first, where there will certainly be space for it. You can transfer it elsewhere later, by floppy, SSH/SFTP or web.
4. Calculate the expected values of the amplitude ratio and phase difference at each frequency, and compare them to their measured values.
5. Do a non-linear least squares fit to your data to find the conductivities. How different are your values from the accepted values?
6. For each metal, compare the results obtained by solving each data point for a skin depth value versus the expected results using your average σ found above and the expected σ .